Forecasting German Electricity Demand, 2015-2020

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Github Link: https://github.com/Saiprathap22/Forecasting-German-Electricity-Demand-

1. Introduction

Electricity cannot be effectively stored, which means the need for constant balance of electricity supply and demand, and forecasting mistakes cost money in both cases. The appropriate time horizon of forecasting defines the appropriate methods of forecasting; intraday dispatch needs very short-term forecasts, while capacity planning needs forecasting horizons from several months to several years, when seasonality and weather are dominant over dynamic factors.

In this work, the national load of Germany is modeled with data from the Open Power System Data (OPSD, 2020) dataset and the forecasts are made for the last two years of the dataset. The goal here is not the optimal minimization of the absolute error, but understanding the reason why certain model types perform good or bad for the two-year horizon and whether increased complexity of regression variables, ensemble trees, and recurrent neural networks improve accuracy. The benchmark is seasonal naive method, because according to the forecasting literature, the method is justified by its ability to beat the free-of-charge benchmark (Hyndman and Athanasopoulos, 2021).

2. Data and Preparation

The load data used in this analysis were taken from the OPSD 60-minute time series (release 2020-10-06), namely DE_load_actual_entsoe_transparency, that is, the actual load in Germany in MW, published on the ENTSO-E Transparency Platform. The dataset was extracted starting from 1 January 2015 until 1 October 2020 and includes 50,401 records, all of them were converted to GW for convenience. The temperature data were downloaded from the Open-Meteo ERA5 archive in the format of daily mean 2m temperature in Berlin (52.52°N, 13.41°E) as a single proxy of German temperature.

There are two important preparation issues that influence the results significantly and should be explicitly mentioned since they are not visible mistakes but silent fails. There is one missing value in the record, which is located in the first row. The linear interpolation cannot work for missing data in the first row, so the backfill approach is applied to it. If this missing value is not corrected, it will be safely ignored in Parts 1-5 during the weekly resample but will destroy each LSTM prediction in Part 6 because of propagation in the StandardScaler used in the neural network.

Secondly, the hourly time series is aggregated to weekly means. Incomplete initial and end calendar weeks (containing 95 and 74 hours correspondingly) are excluded. However, the two weekly transition weeks for every year consist of 167 or 169 hours, and not 168, and the filtering out of weeks having only 168 hours leaves 11 valid weeks out of the data. It results in gaps appearing in the weekly index, which violates the requirement on equidistant index for SARIMA model: in one of the iterations of the experiment, the misalignment of index caused the error of the seasonal-naive model to grow from 2.99 GW to 4.85 GW, and the value of STL seasonal strength to drop from 0.884 to 0.572, just because of the index misalignment. Therefore, the proper threshold is 167 hours or more.

The obtained weekly time series has 299 observations from 11 January 2015 to 27 September 2020, its mean value is 55.51 GW, and the maximum and minimum values are 63.60 GW and 46.50 GW correspondingly. 104 latest weeks (from 7 October 2018 to 27 September 2020) are left aside as a test set for all models. Thus, the training set has 195 weeks.

3. Exploratory Analysis and Stationarity

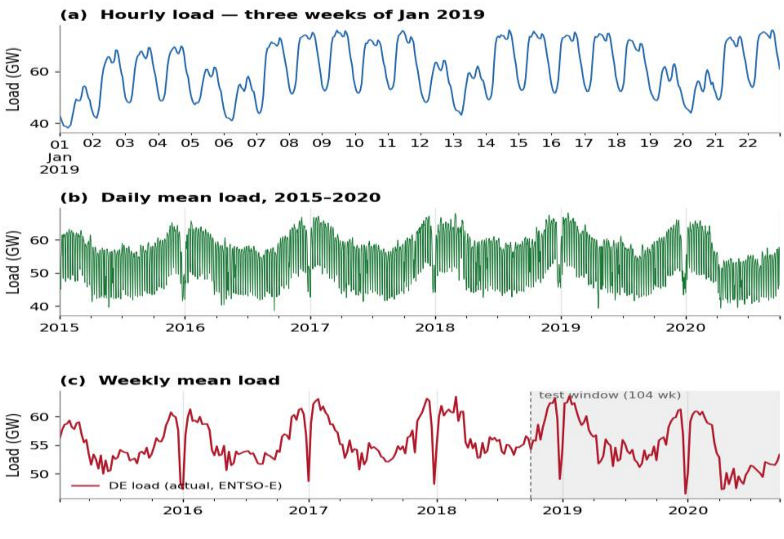


Figure 1: German load at three resolutions. (a) hourly, three weeks of January 2019; (b) daily means; (c) weekly means, with the 104-week test window shaded.

Three distinct seasonal patterns of a hierarchical nature are described in Figure 1. At an hourly resolution level, the most significant seasonality is a daily seasonality pattern, which is associated with low power consumption values overnight (approximately 45 GW), as well as a high value during midday (approximately 70 GW). Another type of seasonality is a weekly one, with the level of consumption on weekends being 15% lower compared to weekdays due to the suspension of industrial and commercial activities. However, at a weekly aggregation level, the daily and weekly patterns become smoothed out, providing dominance of annual seasonality – that is, consumption in winter exceeds summer consumption due to heating needs and fewer daylight hours in Germany. There are two annual sharp negative peaks that occur during Christmas and Easter periods; hence the covariate public holiday is used later on.

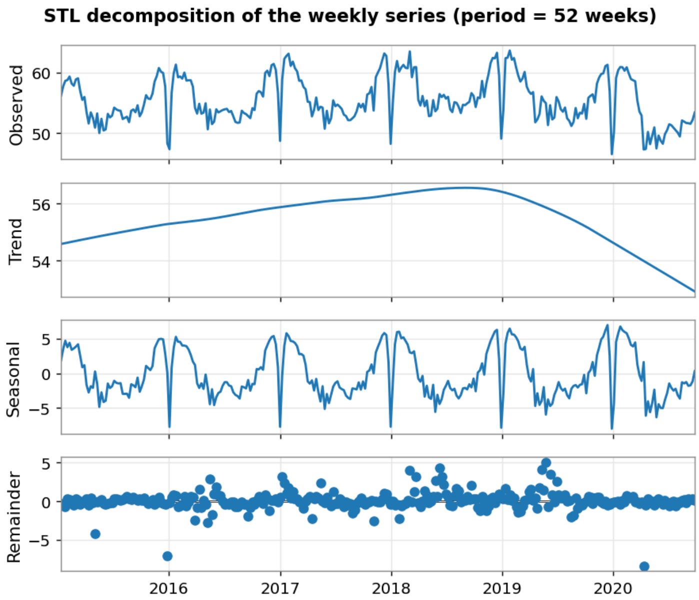


Figure 2: STL decomposition of the weekly series with a 52-week period.

The decomposition of the STL for the weekly time series presented in Figure 2 decomposes the series into trend, annual seasonality, and remainder components. Seasonality is strong and very regular, whereas trend is weak and slightly decreasing, with a noticeable inflection at the end of the series. Measured using the variance-based strength criteria of Hyndman and Athanasopoulos (2021), seasonal strength and trend strength amount to 0.884 and 0.437, respectively. In conclusion, seasonality dominates in this time series together with a slow trend; thus, the lack of a seasonal component in any time series model will lead to poor forecasting results.

Table 1: Stationarity tests on the weekly series

Transformation	ADF p	KPSS p	Interpretation
Level	0.0021	0.1000	Both tests indicate stationarity
First difference (d = 1)	0.0000	0.1000	Stationary
Seasonal difference (D = 1, s = 52)	0.3795	0.0100	Both tests indicate non-stationarity
Seasonal and first difference	**0.0000**	**0.1000**	**Stationary**

Accordingly, the null hypothesis of the ADF is the unit root, and the null hypothesis of the KPSS is that the series is stationary. We should emphasize here that KPSS p-values are capped at 0.10 and 0.01.

In Table 1, we find the following contradictory result that requires further explanation. The Augmented Dickey–Fuller test rejects the unit root for the raw level series ($p = 0.002$) but the KPSS test cannot reject stationarity ($p = 0.10$). Naively, one might conclude that the series does not need differencing. But the ACF plot in Figure 3 shows that the decay is slow and there is a large spike at lag 52. The answer lies in the understanding that ADF and KPSS do not test for seasonality, but for the existence of a stochastic trend: the very strongly seasonal series without unit root may pass both tests but be inappropriate for non-seasonal model. Tests are diagnostics, not an alternative for checking the correlogram.

Another curious thing about this example is that seasonal differencing alone makes the situation even worse (ADF $p = 0.38$). Using only 195 training weeks, seasonal differencing at lag 52 results in almost a year worth of data lost and the residual part of the short over-differenced series where tests are not powerful enough. And the reason is that only the combination of first and seasonal differencing produces unambiguously stationary series. This combination is chosen in our SARIMA models.

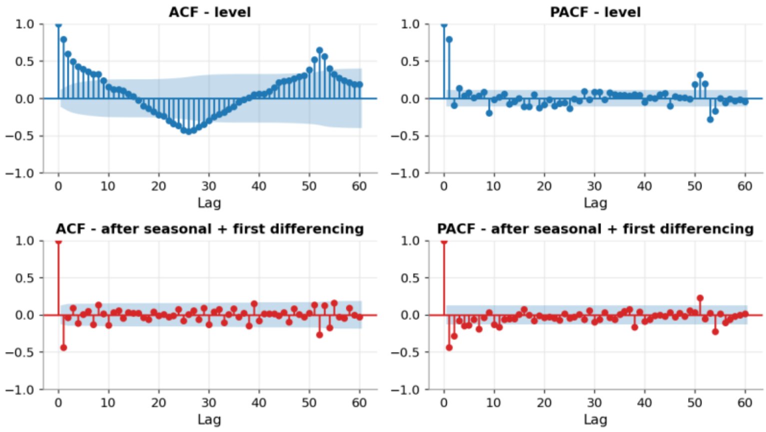
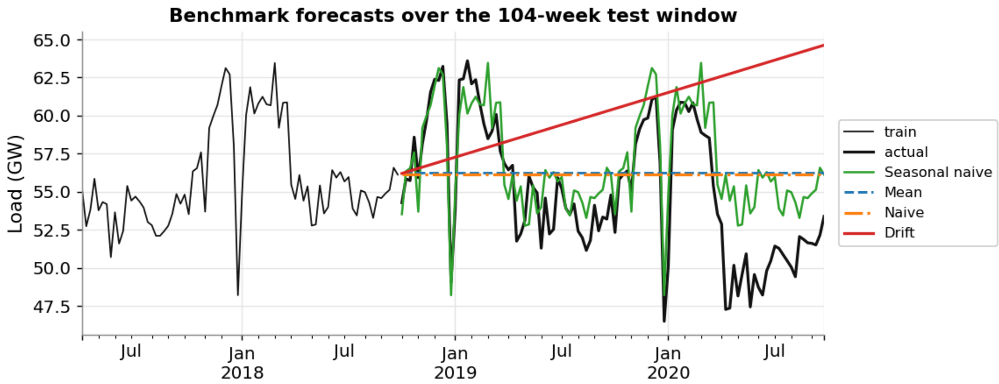


Figure 3: ACF and PACF of the weekly series before and after seasonal and first differencing.

4. Benchmark models

Benchmark models are fitted and employed for predicting the 104-week test window: the training mean, the naive model which predicts using the last observation, the seasonal naive method that uses the value of the same week of last year, and the drift method which predicts using a straight line with the drift slope being equal to the average week-to-week increase. As required by the task, benchmark models are trained using the data from 2017 onwards (92 weeks), while statistical and machine learning models will be trained using the entire training data (2015 onwards) in further parts of the task. Since the drift slope is estimated based on 2017-2018 where there was a notable increase in load, the drift line is highly inclined over the horizon.



RMSEs of the seasonal-naive forecast are 2.99 GW versus 4.51 and 4.48 GW in case of the mean and naive forecasts, respectively (the improvement is one-third compared to the competitors) and 8.01 GW for the drift forecast. There is no need for any statistical explanations of such results: the problem is structural, not statistical. In the series where there is an approximate annual variation of ±10 GW, the naive and mean forecast cannot take into account the key characteristic of the time series. The most disappointing is the performance of the drift forecast because the projection of the upward trend of 2017-2018 into future will put us even more far from the series when it goes down due to structural break in 2020.

5. SARIMA

A SARIMA model is specified in light of the seasonality of the series. Each of the 147 possible values of p from 0 to 6, d from 0 to 2, and q from 0 to 6 has been estimated using maximum likelihood estimation in statsmodels, assuming the seasonal order to be the airline type (1,1,1,52) and sorted based on their AIC; later, the seasonal order was optimized for (P,D,Q) = $\{0,1\}^3$ given the non-seasonal order that has been selected. An exhaustive optimization would involve fitting 1,176 state space models at a 52-week seasonal cycle.

Table 2: Five best (p,d,q) orders by AIC, seasonal order fixed at (1,1,1,52)

Order (p,d,q)	AIC	BIC
(1,1,6)	**297.57**	**321.76**
(0,1,6)	299.10	320.87
(2,1,6)	299.51	326.11
(3,1,6)	299.63	328.65
(1,0,6)	301.47	325.78

SARIMA(1,1,6)(1,1,1,52) model was selected having AIC value of 297.54. There is one methodology-related concern that needs to be raised. Grid changes differencing parameter d, while AIC values calculated for time series differenced at different p are calculated for different datasets and cannot be compared directly (Hyndman and Athanasopoulos, 2021). While loop over d parameter was done according to specification, the decision to justify differencing should be based on the information from Table 1 more than on AIC values. The fact that four models with lowest AIC values have d = 1 should be noted as well.

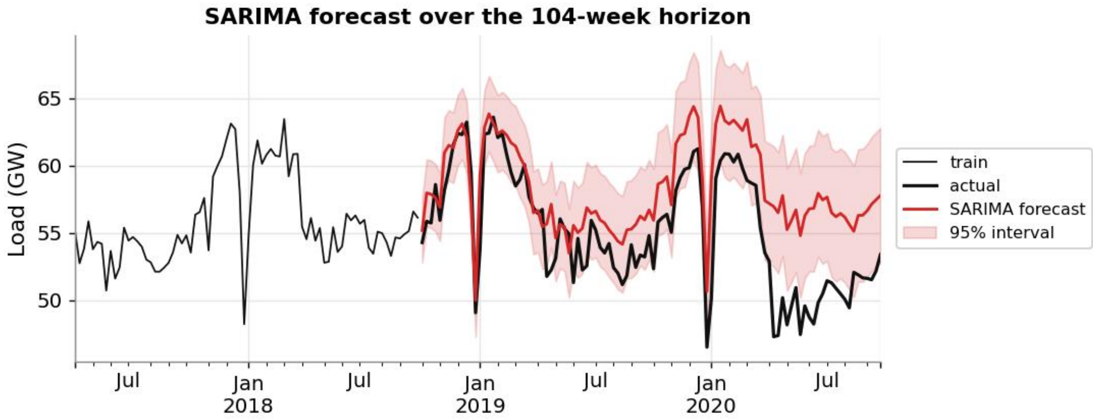


Figure 5: SARIMA(1,1,6)(1,1,1,52) forecast with 95% prediction intervals.

The fitted model produces RMSE = 4.03 GW, MAE = 3.30 GW, and MAPE = 6.26% on the test window. Hence, its out-of-sample performance is significantly worse than the benchmark seasonal naïve model, by 35% in RMSE. This is the main negative finding of this research and should be explained rather than concealed.

There are three reasons for it. First, the selection criterion is not appropriate to the problem. The AIC penalizes in-sample log-likelihood of the model, which is mainly determined by one-step-ahead prediction for the ARIMA process; however, the forecast horizon spans 104 periods ahead. After several weeks, the AR and MA components exponentially fall down to zero, thus leaving the forecast to tend to the seasonally differenced drift. In consequence, the six MA terms which received high weightings under AIC virtually contribute nothing at the horizon of interest but add to the estimation variance. Second, the sample size is too short – 195 training periods correspond to less than four annual cycles, thus making the seasonal term to be estimated based on only four repeats of the pattern. Third, a structural break, discussed in Section 8, makes the forecast even harder.

6 SARIMAX: Conditioning on Temperature and Holidays

The physical nature of electricity demand is determined by the weather, hence the inclusion of temperature as an exogenous variable. Daily average temperatures in Berlin are grouped into weekly average temperatures, minima, and maxima, and are converted into heating degree days and cooling degree days, at a base temperature of 15.5 °C for heating degree days and 22.0 °C for cooling degree days. The concept of degree days incorporates the non-linear nature of the electricity demand-temperature relationship as follows: electricity demand increases with decrease in temperature below a comfort level and vice versa.

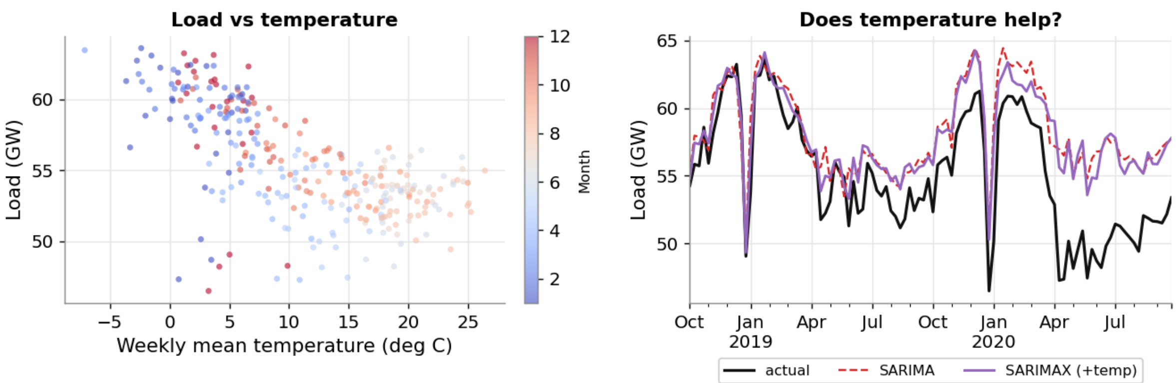


Figure 7: Left — weekly load versus Berlin's weekly mean temperature, classified by month. Right — Conditional forecast for the SARIMAX model vs. SARIMA.

As seen from the scatterplot in Figure 7, there is the expected highly significant negative correlation: the heating degree days correlate positively (+0.698) with the weekly load and negatively (−0.653) with the mean temperature. Including these regressors in the selected SARIMA significantly lowers AIC from 297.5 to 241.5 and RMSE from 4.03 to 3.67 GW.

Table 3: Estimated exogenous coefficients

Regressor	Coefficient	Interpretation
temp_mean	+0.353	Sign inverted by collinearity
heating_degree	**+0.082**	**Colder weeks raise demand**
cooling_degree	−0.070	Not significant
temp_lag1	−0.010	No weekly-scale thermal lag
holiday_days	**−1.809**	**Each holiday day removes ~1.8 GW**

Table 3 requires careful reading, and one specific coefficient deserves special attention. Mean temperature has a negative correlation of −0.65 with load but is estimated as having a positive coefficient. This is a typical example of multicollinearity because degree days are a deterministic almost linear function of mean temperature on the part of its domain when heat is required, and the two regressors compete in explaining the same variation. As a result, their coefficients become uninterpretable, even though both of them together carry the right information. The proper solution would be to keep the regression specification using degree days and drop the mean temperature as a covariate. The model was used in its initial form only to reveal diagnostics instead of hiding them; nevertheless, the coefficient on temp_ mean must not be interpreted as an elasticity.

Insignificant cooling degree days have nothing to do with methodology and are explained by the specifics of Germany; the number of houses with air conditioning is relatively low, and summer cooling does not contribute much to the total load in the country but will dominate in Spain and Italy. The coefficient on holiday is big and physically reasonable.

There are, however, two nuances to the seemingly improved model. First, SARIMAX remains worse than the seasonal naive benchmark by 23%. The difference between the significant in-sample improvement in likelihood (AIC reduction of 56 points) and the moderate out-of-sample accuracy improvement demonstrates the reason why information criteria cannot be used to evaluate forecasting performance. Second, coverage of the prediction interval falls from 70.2% to 40.4%. Introducing additional regressors led to the reduction of the estimated variance and hence narrowing the intervals, but the actual errors are driven by the pandemic, which cannot be explained by any temperature series. The model became more confident without becoming more accurate, which is useless for a system operator who needs to size up reserves.

Finally, a term explanation to interpret the results properly. As this test-period temperature is the observed value and not a forecast of weather, it is a conditional/explanatory forecast, not an operational one. The accuracy of this forecast represents the upper bound of what could be reached in practice.

7 Feature-Based and Neural Models

7.1 Random Forest and Gradient Boosting

Fifteen features are constructed: weekly mean, minimum, and maximum temperature, heating and cooling degree days, temperature lags of 1 and 2 weeks, 2 Fourier components for week-of-year, a linear time index, and holiday counts and indicators. Load lags are explicitly left out because at the 104-week forecast horizon, there is no way to derive a lagged load at the forecast origin without recursive forecasting, which would involve compounding errors. Excluding the load lag ensures that the forecast is a multi-step one and places the tree-based models in the category of conditional forecasts alongside SARIMAX.

Each model is exhaustively grid searched through 36 possible parameter combinations and evaluated by the cross-validated RMSE. Standard K-fold cross-validation is not appropriate in this case, since shuffling of folds might lead to information leakage due to training on weeks coming after the weeks being validated. TimeSeriesSplit is used instead, employing four expanding folds where validation periods always come after the training periods. For Random Forest, 500 trees are used with maximum depth of 8 and full feature set at each split (CV RMSE = 1.866). For Gradient Boosting, 500 stages are used with maximum depth of 2, learning rate of 0.05, and 0.8 subsampling (CV RMSE = 1.568). Depth of the trees in the boosted model speaks volumes about itself; the search did not find any deeper trees useful given only 193 training samples.

Table 4: Tuned hyper-parameters (TimeSeriesSplit, 4 folds, 36 candidates each)

Model	Selected parameters	CV RMSE (GW)
Random Forest	500 trees, max_depth 8, min_samples_leaf 1, max_features 1.0	1.874
Gradient Boosting	500 stages, max_depth 2, learning_rate 0.02, subsample 0.8	1.606

The RMSE for the Random Forest model is 2.846 GW, while that for the Gradient Boosting model is 2.881 GW. The two models have done better than the seasonal naive method by 4.9% and 3.7% and are the only two methods in the research that have surpassed the benchmark.

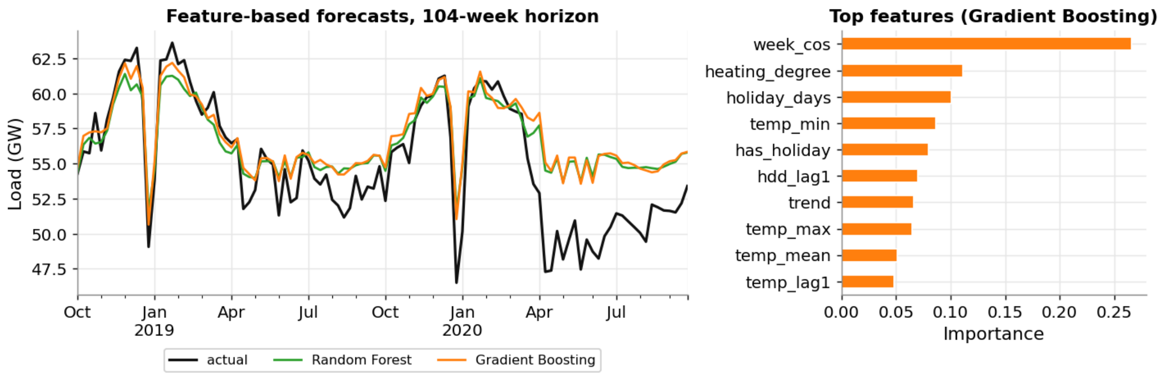


Figure 8: Predictions made using Random Forest and Gradient Boosting methods, with feature importances plotted for Gradient Boosting.

The exact reason behind this should be clearly explained since it differs from the case with SARIMA. Feature-based models incorporate seasonality in a deterministic fashion via calendar-based features, such as Fourier terms of week-of-year, rather than modeling seasonality as stochastic process and predicting its future behavior. Deterministic annual seasonality does not decay in strength at greater horizons; thus, week 104 is predicted just as well as week 1. At the same time, SARIMA model has to propagate an estimated structure of seasonal ARMA process for 104 time steps, with uncertainty of its estimations accumulating with horizon.

The feature importances should also be interpreted with caution. For example, week_cos alone explains 27% of the Gradient Boosting model, with heating degree-days taking 12% and number of holidays accounting for 10%. Thus, these tree ensembles act as a fairly advanced version of seasonal naive model, which happens to take into account additional information about weather and holidays. This way, the improvement of 5% can be viewed in different light and a more reasonable conclusion can be drawn, other than the general one about superiority. One should also note the structural drawback of decision trees; namely, they cannot predict values outside of the target variable range observed during training.

7.2 LSTM on Hourly Data

Recurrent Neural Networks, and Long Short-Term Memory in particular (Hochreiter & Schmidhuber, 1997), are widely used in the area of load forecasting thanks to their gate functions that allow for gradients to flow through long sequences without vanishing, making the network capable of remembering information from many time steps back. Good results in short-term load forecasting for residential loads have been achieved using LSTMs by Kong et al. (2019), while Bedi and Toshniwal (2019) show similar results on an aggregated level. Contrasting the above-mentioned studies, Makridakis et al. (2018) offer a study in which machine learning methods do not outperform well-chosen statistical models and more complicated models do not necessarily perform better.

The inputs to the network include a 48 hours window of the scaled load along with cyclic encodings of hour, day of week, and month in sine–cosine functions, along with a variable for whether it is a weekend and one for whether it is a public holiday in Germany – altogether nine channels. Cyclic encoding guarantees that hour 23 and hour 0 are neighbors in terms of features, not farthest from each other. The scaler is fit only on training rows. Four architectures have been tested on a validation split containing last 26 weeks of the training data, which never overlap with the test data.

Table 5: LSTM architecture search (validation RMSE, GW)

Layers (units)	Dropout	Learning rate	Validation RMSE
[32]	0.0	1e-3	1.167
[64]	**0.1**	**1e-3**	**1.024**
[64, 32]	0.1	1e-3	1.091
[32]	0.2	5e-4	1.465

A sole 64-unit layer with a dropout rate of 0.1 was chosen. The notable thing here is that the combination of [64, 32] showed slightly worse results, and excessive dropout was also harmful. Having about 28,500 training windows with a very structured target variable, the increase in capacity leads to increased variance without any reduction in bias.

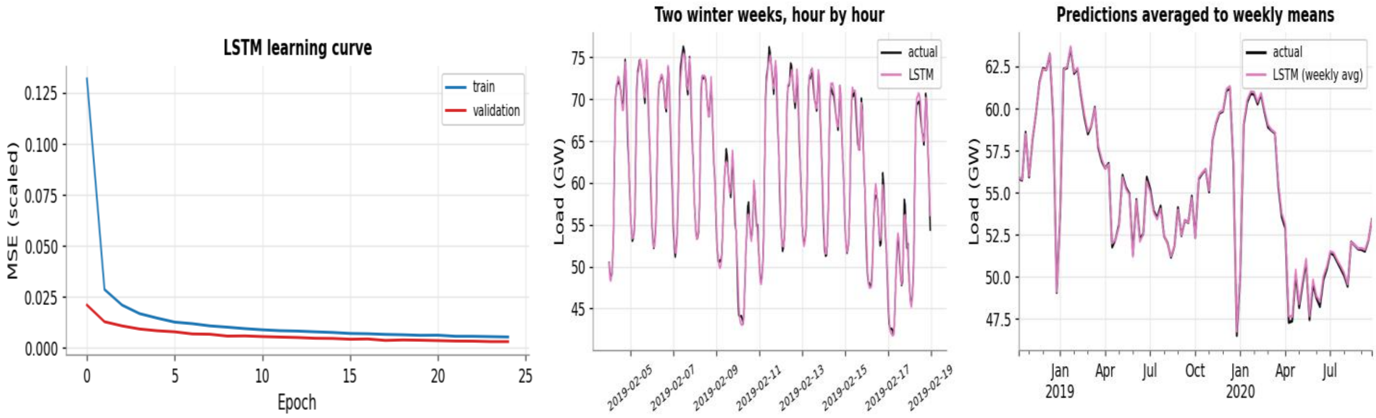


Figure 9: LSTM learning trajectory, two weeks of one-hour ahead predictions, and week-average predictions.

In one-step-ahead walk-forward evaluation with frozen weights for the entire two years' test period, the LSTM achieves hourly RMSE of 0.672 GW and MAPE of 0.95%; the weekly average RMSE reduces to 0.142 GW. Figure 9 depicts the ability of the network to learn the daily and weekly cycles, as well as holidays' troughs, with great accuracy.

This performance cannot be compared to the weekly models' performance naively, and the temptation to declare LSTM a better approach should not arise. For prediction of hour t, the network uses the observed true load up to hour t−1. The weekly models predict 104 weeks ahead, knowing nothing beyond the forecast origin. A comparison of RMSE 0.13 GW vs. 2.99 GW is a difference in information content, rather than model performance. A naive seasonal forecasting method assuming one-hour ahead load information ("next hour equals this hour") would work just fine. What the LSTM performance shows is that load has high predictability at short time horizons – this is where such models are used for intraday and day-ahead dispatch, not for two years ahead.

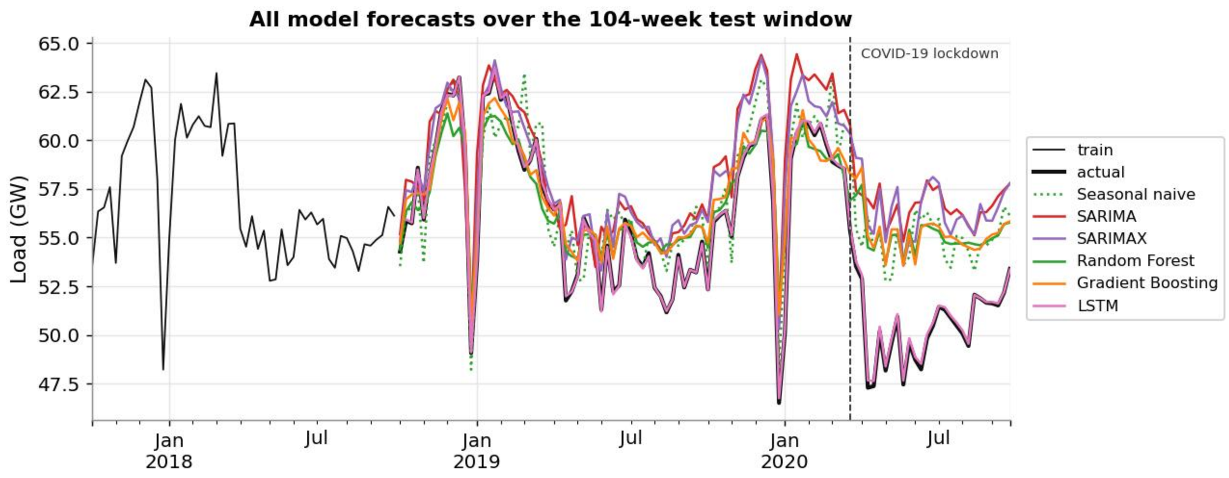
To demonstrate this fact explicitly, the network was evaluated as a real recursive forecast: rolled forward for two weeks without using any observed data, using only its own previous predictions as inputs, while keeping all the calendar features known, for six windows spread across the test period. The averaged RMSE for such a forecast grows to 2.86 GW, vs. 0.67 GW for one-hour-ahead prediction – three times higher once the network does not have true recent data anymore. This is the right comparison to make for weekly models. In this light, LSTM is not better now, and the original headline performance number is mostly determined by the forecast horizon.

8 Results and Critical Discussion

Table 6: Forecast accuracy over the 104-week test window

Model	RMSE (GW)	MAE (GW)	MAPE (%)	Skill vs SN (%)
Mean	4.513	3.894	7.24	−50.9
Naive	4.476	3.857	7.15	−49.6
Drift	8.011	6.514	12.45	−167.8
Seasonal naive	**2.991**	**2.291**	**4.36**	**0.0**
SARIMA (1,1,6)(1,1,1,52)	4.029	3.298	6.26	−34.7
SARIMAX (+ temperature)	3.675	2.928	5.58	−22.7
Gradient Boosting	2.881	2.147	4.14	+3.7
Random Forest	**2.846**	**2.154**	**4.14**	**+4.9**
LSTM — weekly aggregate	0.133	0.108	0.20	+95.5

All rows marked as "benchmark" are trained with data starting in 2017 according to the assignment; all other models are trained using the whole 2015 training set; thus, benchmark and model rows have been trained on different periods on purpose. LSTM row cannot be compared to the previous rows directly because this row makes a forecast for the next hour using the load value up to time t −1, while all other rows make forecasts for 104 weeks starting from a constant origin point. Performance metrics for raw hours: RMSE 0.672 GW, MAPE 0.95%.



All forecasts generated by models in the 104-week testing window are shown in Figure 10; the dotted line marks the March 2020 lockdown.

8.1 The COVID-19 Structural Break

Figure 10 shows one distinct feature of the test window, which consists in the fact that starting from March 2020, all forecasts remain persistently much higher than the observed data. The last 28 weeks of the 104-week test window fall into the period after the first national lockdown. In Q2 2020, the average weekly load of Germany was 49.28 GW, whereas in the same weeks of 2019 it stood at 54.09 GW, which is 8.9% lower. The reason for the reduction is not weather but the shutdown of industries and commercial facilities. Bahmanyar et al. (2020) show a similar effect across several European systems and correlate its size with containment measures.

Table 7: Test error split at the March 2020 lockdown (76 weeks before, 28 weeks after)

Model	RMSE pre-COVID	RMSE COVID	Skill pre-COVID (%)
Seasonal naive	1.887	4.855	0.0
SARIMA	2.699	6.364	−43.1
SARIMAX (+ temperature)	2.195	6.090	−16.3
Random Forest	**1.648**	**4.767**	**+12.7**
Gradient Boosting	**1.568**	**4.914**	**+16.9**

Table 7 changes the story told by Table 6 considerably. Over the 76 weeks before the outbreak started, both tree ensemble models outperform the seasonal naïve model by about 12–13%, thus almost tripling the difference relative to the entire sample. Furthermore, the difference in every model increases threefold after the introduction of the lockdown measures. Therefore, the comparison based on the entire window underestimates the performance of every model since almost one-quarter of the entire sample period takes place in a regime that the model built during 2015-2018 cannot forecast. It can be concluded that ranking of a model by a structural break tests the robustness of the model at least as much as its accuracy.

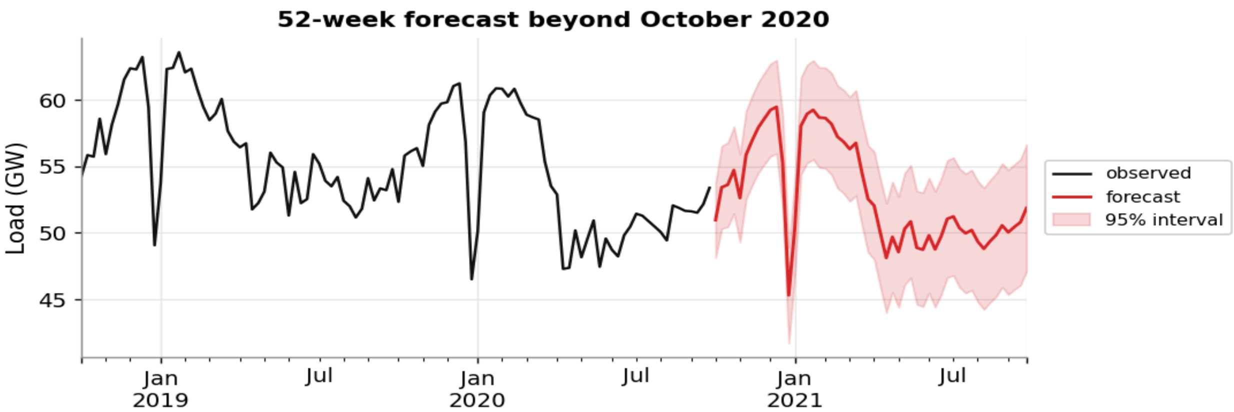


Figure 11: Re-estimated SARIMA model for full data set with a 52-week forecast beyond October 2020 and 95% prediction intervals.

In Figure 11, an out-of-sample forecast using the re-estimated SARIMA model for the full 299 periods with a 52-week forecast beyond the last observation is shown. The prediction intervals for this case are somewhat large, but even taking into account the coverage results stated above, it is probably the case that they are still not sufficiently wide. Any implementation of the forecast on practical grounds should be based on the assumption that the pandemic period is either going to continue or end.

9 Analytical Questions

Which models meaningfully outperform the seasonal-naive benchmark?

The tree models outperform the benchmark in full: Random Forest outperforms by 4.9%, while Gradient Boosting is ahead by 3.7%. SARIMA (−34.7%) and SARIMAX (−22.8%) underperform the benchmark. Looking at the 76-week sub-sample excluding the pandemic weeks, the tree ensembles outperform the benchmark by approximately 13-17%, which represents the fairest assessment of the models' capabilities in the regime they were built on. Whether the improvement in RMSE of 4-5% over 104 time periods counts as meaningful is a matter of opinion: a statistical test has not been applied to confirm it, and a Diebold-Mariano test would be needed to answer the question. It is acknowledged as a limitation of this analysis, but not justified otherwise.

How was data leakage prevented in construction of temperature lag features?

Temperature lags use a positive shift that is inherently looking into past data, hence, the value of a lag k in week t is the temperature of week $t-k$, and no future data is included. The shift is applied before the split, so any resulting NaNs will occur in the beginning of the training data and be dropped in it, not in the test. None of the features is derived from the load data, so no target lag is ever leaking information past the split. The standardisation of the neural model uses training data rows only, and the tree models are optimised with TimeSeriesSplit CV that never validates on data that occurred prior to training. The LSTM uses the final 26 weeks of training data as validation and never sees test data during architecture selection.

Justify the differencing orders and the seasonal period.

The seasonal period of $s = 52$ stems from the weekly sampling and annual seasonality that is corroborated by the spike in ACF at lag 52 (Figure 3) and a seasonal strength of STL of 0.884. The first differencing order of $d = 1$ is chosen based on the AIC grid, since the four best models have selected it independently. It is further justified by Table 1, where the combination of $d = 1$ results in ADF $p < 0.001$ and failure of KPSS to reject stationarity. Seasonal differencing ($D = 1$) eliminates the annual seasonality and Table 1 shows that it is not sufficient independently (ADF $p = 0.38$) and that $d = 1$ and $D = 1$ together lead to stationarity. As mentioned in Section 5, AIC values cannot be directly compared across differencing orders, so the stationarity evidence is more important.

Do temperature and holiday covariates improve accuracy, and are they known at the forecast origin?

Yes, they increase accuracy: RMSE decreases from 4.029 to 3.675 GW and AIC from 297.5 to 241.5, holiday count (−1.81 GW per holiday day) and heating degree-days being statistically significant at $p < 0.001$. The improvement is not large enough to beat the benchmark and cooling degree days are not significant.

On availability, the two covariates differ greatly. Public holidays are calendar determined and thus known years in advance. Temperature is not: weather forecasting becomes increasingly unreliable after a fortnight and the 104-week path of temperature used in the analysis is observed, not predicted. Hence, the SARIMAX and tree models give conditional or explanatory forecasts and represent an optimistic upper bound on operational performance. An operational forecast would replace it with climatological normals, losing most of the 9% gain.

Interpretability vs. complexity of SARIMAX, feature-based and neural models.

SARIMAX fits roughly fourteen parameters, each with a sign, a standard error, and a p-value, and provides prediction intervals; it is the most interpretable of all and the only model that provides uncertainty quantification on its own. Tree ensembles train 500 trees in seconds, provide understandable and physically meaningful feature importance rankings, but lack signs, interaction structure, and intervals. The LSTM has roughly nineteen thousand weights, needs to be scaled, windowed and searched for optimal architecture, and is essentially uninterpretable.

The traditional trade-off of interpretability vs. accuracy is hard to reconcile with these findings. The most interpretable model, SARIMAX, is among the least accurate models for this horizon, and the most transparent method possible – seasonal naive – is one of the most robust. In reality, there is a trade-off between interpretability and robustness on one hand and marginal accuracy on the other hand, and the margin is small.

Which model would you recommend for operational use?

For a two-year planning horizon that we have analyzed, Random Forest is recommended, with seasonal naive maintained as a permanent fall-back and sanity check. In terms of accuracy, it is the only model to beat the benchmark for the whole period (+4.9%), and beats it by 12.4% in the pre-pandemic regime. In terms of maintenance costs, it is the most inexpensive serious model in the study: seconds to retrain, no order identification, no special-purpose hardware required. In terms of interpretability, it is good enough, since its importances are understandable and drivers physically meaningful. It is also the worst in terms of uncertainty, since it provides no intervals at all; quantile regression forests or a bootstrap ensemble can fix that and must be seen as a prerequisite of deployment rather than an option.

There are two caveats in this recommendation. First, it relies on a temperature variable not available on a two-year horizon and in production will use climatological normals, thus losing much of its advantages. Second, under a structural break it is no better than the free benchmark. Therefore, the model for operational use is layered: Random Forest for planning horizons using climatological temperatures, seasonal naive as a permanent fall-back that the complex model has to prove its superiority over, and LSTM as the tool specifically designed for the dispatch problem, where its advantage is real and significant.

10 Limitations and Future Work

The biggest limitation is that all metrics in this report rely on a single 104-week evaluation window, and this window just happened to contain a pandemic. Rolling origin cross-validation (Bergmeir and Benítez, 2012) would assign error bars to every figure in Table 6 and show how much of the model ranking was signal and how much was random chance associated with one specific window. In connection with that, no formal test of forecast accuracy differences was conducted; the Diebold-Mariano test is a standard procedure and must accompany any comparison of different models' forecasts.

The evaluation used in this report is purely point-based; the RMSE measure says nothing about prediction intervals, while the failure of interval calibration documented here (70.2% and 40.4% of coverage against nominal 95%) can be considered the most important failure found. Appropriate scoring rules like pinball loss or CRPS (Hong and Fan, 2016) should become the main criterion.

Regarding the data, Berlin is a poor approximation of the national temperature for Germany; a population-weighted average of temperature across German regions would be a strict improvement at zero extra cost. The collinearity of mean temperature and heating degree-days can be resolved by keeping only the latter. Finally, the LSTM has been evaluated only one-step ahead; its multi-step or sequence-to-sequence alternative would give the LSTM a chance to compete in the forecasting task, and regime-switching or intervention specification would help the statistical models cope with the COVID shock.

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